

Samanvay — Final System Architecture (MVP + Growth Path)

Vision

Samanvay is a coordination operating system for the social sector.

It enables NGOs, volunteers, donors, institutions, CSR partners, and government bodies to collaborate through a shared coordination infrastructure while retaining ownership of their own resources.

The platform does not own NGO resources.

The platform owns the coordination layer.

Core Philosophy

Every participant has autonomy.

Every collaboration requires consent.

Every action is auditable.

Every allocation is traceable.

Every impact is verifiable.

Ownership Model

Every allocatable entity belongs to one of three domains.

Private

Owned by a specific organization.

Examples:

- Inventory
- Funds
- Internal volunteers

- Assets

Rules:

- Only owner organization can modify.
 - Other organizations cannot access directly.
 - Can participate in coordination workflows only through explicit approval.
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Shared

Owned by Samanvay.

Examples:

- Community volunteer pool
- Community emergency fund
- Shared logistics partners
- Knowledge resources

Rules:

- Accessible through platform governance.
 - Allocation controlled by platform policies.
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Collaborative

Temporary shared ownership.

Examples:

- Joint events
- Joint campaigns
- Cross-NGO volunteer assignments
- Resource transfers in progress

Rules:

- Requires approval from all participating parties.
 - Expires after completion.
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System Architecture

Samanvay consists of three layers.

Layer 1: Foundation Layer Layer 2: Coordination Engines Layer 3: Intelligence & Governance

Layer 1 — Foundation Layer

Identity System

Entities:

- User
- Organization
- Membership
- Role

Roles:

- Admin
- Coordinator
- Member
- Volunteer
- Donor
- Verifier
- Platform Admin

Purpose:

Common authentication and authorization system for every module.

Organization Network

Purpose:

Create a network graph between organizations.

Features:

- NGO registration
- Verification
- Partnerships

- Organization profiles
- Sector tagging
- Geographic coverage

Future:

- Trust score
 - Organization discovery
 - Recommendations
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Event & Campaign System

Purpose:

Operational center of the platform.

Examples:

- Food distribution
- Medical camp
- Awareness campaign
- Fundraiser
- Emergency response

Events consume:

- Resources
- Volunteers
- Funds
- Capabilities

Events produce:

- Reports
 - Verification
 - Impact
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Layer 2 — Coordination Engines

These are the core differentiators of Samanvay.

Engine 1 — Resource Coordination Engine

Purpose:

Coordinate physical resources between organizations.

Examples:

- Food
- Medicine
- Equipment
- Vehicles
- Clothes
- Shelter capacity

Core Workflow:

Need → Discovery → Offer → Approval → Transfer → Completion

Rules:

- No automatic inventory access.
- Only suggestions and offers.
- Transfer requires explicit approval.

Core Models:

- Resource
- ResourceNeed
- ResourceOffer
- Transfer

Future:

- Reservations
- Shared resource pools
- Expiry tracking
- Resource forecasting

Engine 2 — Volunteer Coordination Engine

Purpose:

Coordinate human resources.

Volunteer Attributes:

- Skills
- Availability
- Location
- Languages
- Certifications
- Experience

Workflow:

Need → Matching → Invitation → Acceptance → Assignment → Attendance

Rules:

- Volunteer consent required.
- NGO cannot force assignments.

Core Models:

- Volunteer
- VolunteerNeed
- VolunteerAssignment

Future:

- Volunteer trust score
- Certification system
- Volunteer passport
- Cross-NGO volunteering

Engine 3 — Fund Allocation Engine

Purpose:

Manage and allocate money transparently.

Fund Types:

1. NGO Fund
2. Campaign Fund
3. Community Fund

Workflow:

Contribution → Fund → Budget Allocation → Expenditure → Reporting

Rules:

- Append-only ledger.
- No balance mutation.
- Balance derived from transactions.

Core Models:

- Fund
- FundEntry
- Budget

Future:

- Donor dashboards
- Campaign funding
- Community treasury

Engine 4 — Capability Coordination Engine

Purpose:

Coordinate expertise, not inventory.

Examples:

- Medical response teams
- Legal aid
- Disaster response
- Counseling
- Education programs

Workflow:

Capability Need → Discovery → Offer → Approval → Deployment

This enables organizations to collaborate based on what they can do, not just what they own.

Core Models:

- Capability
 - CapabilityNeed
 - CapabilityOffer
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Layer 3 — Intelligence & Governance

Built after sufficient platform usage exists.

Verification System

Purpose:

Verify that actions actually happened.

Methods:

- OTP
- Image proof
- Document proof
- Third-party verifier
- NGO confirmation

Nothing becomes impact until verified.

Trust System

Purpose:

Measure reliability.

NGO Score Factors:

- Fulfillment rate
- Response time
- Disputes
- Transparency

Volunteer Score Factors:

- Attendance
- Feedback
- Completion rate

Verifier Score Factors:

- Verification accuracy
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Impact Engine

Purpose:

Generate immutable impact records.

Examples:

- Meals distributed
- Beneficiaries reached
- Volunteers mobilized
- Funds utilized

Generated only after verification.

Never edited.

Append-only.

Analytics Engine

Purpose:

Platform-wide visibility.

Examples:

- Resource shortages
 - Volunteer demand
 - Regional trends
 - Event effectiveness
 - Fund utilization
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Governance System

Purpose:

Protect the ecosystem.

Responsibilities:

- Organization verification
- Fraud detection

- Dispute resolution
 - Community fund governance
 - Shared resource governance
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Platform Workflows

Resource Coordination

NGO creates need → Engine finds candidates → Candidate NGO offers help → Requesting NGO accepts → Transfer occurs → Impact generated after verification

Volunteer Coordination

NGO creates volunteer requirement → Matching engine recommends candidates → Invitations sent → Volunteers accept → Event attendance recorded → Impact generated

Fund Allocation

Donations enter fund → Budget allocated to event → Expenses logged → Reports generated → Donor transparency updated

Capability Coordination

NGO requests expertise → Engine identifies capable organizations → Partnership formed → Capability deployed

Long-Term Vision

Samanvay becomes the coordination infrastructure for the social sector.

Organizations remain independent.

Resources remain privately owned.

Volunteers remain autonomous.

Funds remain transparent.

The platform creates coordination, trust, discovery, and collaboration across the ecosystem.

Success is not measured by how many NGOs manage themselves on Samanvay.

Success is measured by how effectively independent organizations can work together through Samanvay.